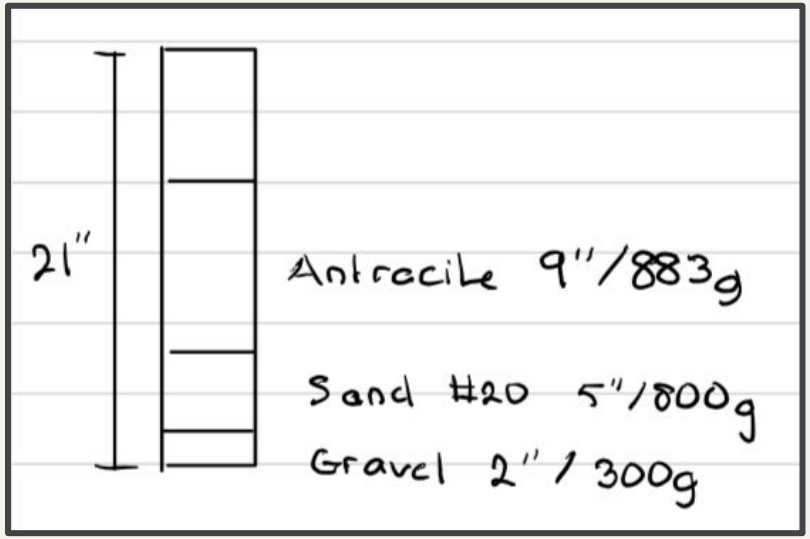
Water Filters & Turbidity

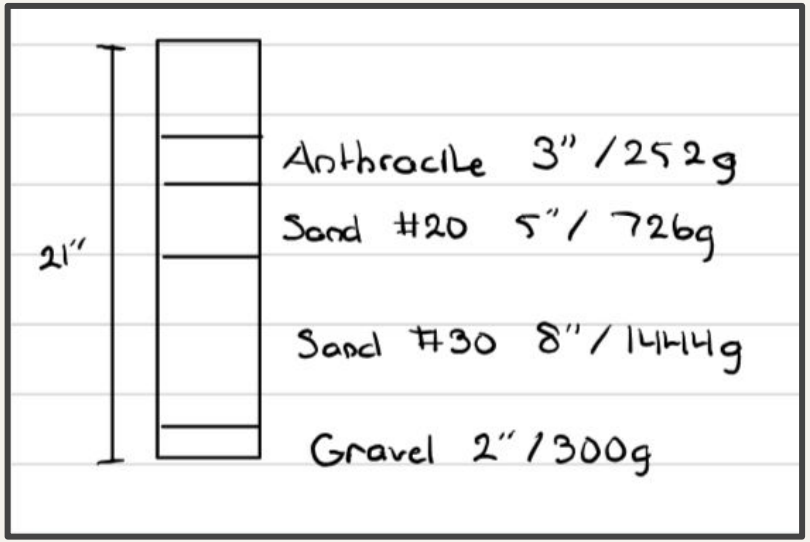
By: Kasin, Nathan, and Tristyn

Filter Design

Filter 1



Filter 2

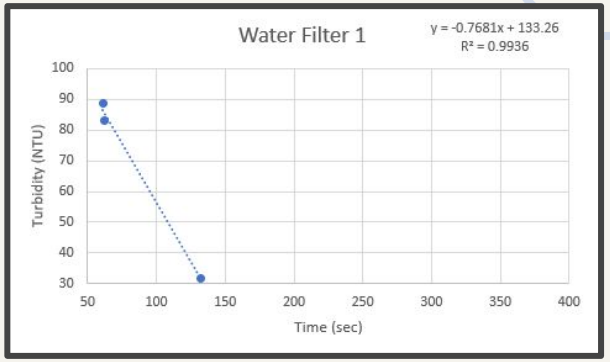


Turbidity Data

Filter 1

Filtrate Turbidity Test #1 (NTU)*	Filtrate Turbidity Test #2 (NTU)	Filtrate Turbidity Test #3 (NTU)	Time to Filter Test #1(min)	Time to Filter Test #2 (min)	Time to Filter Test #3 (min)
31.9	83.1	88.9	2:12.44	1:02.27	1:01.32

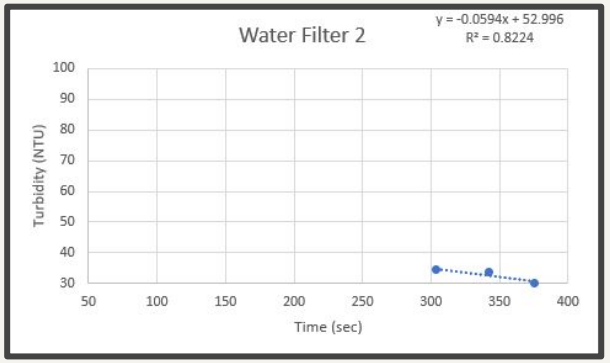
Inches of Anthracite	Inches of Sand (#20)	Inches of Sand (#30)	Inches of Gravel (Coarse Sand)	Total Media Height (inches)
9"	5"	0"	2"	16"



Filter 2

Filtrate Turbidity Test #1 (NTU)*	Filtrate Turbidity Test #2 (NTU)	Filtrate Turbidity Test #3 (NTU)	Time to Filter Test #1(min)	Time to Filter Test #2 (min)	Time to Filter Test #3 (min)
30.1	34.4	33.8	6:15.78	5:04.18	5:42.14

Inches of Anthracite	Inches of Sand (#20)	Inches of Sand (#30)	Inches of Gravel (Coarse Sand)	Total Media Height (inches)
3	5	8	2	18



Improvements

- Increased amount of sand used in filter
  - Longer filter time
- Used both sand types
  - Smaller particle size
- Decreased amount of anthracite
  - More space for sand
  - Less anthracite dust
- Decreased amount of space for backwash
  - Reduce amount poured in at once



Filter #1



Filter #2